

Operating Systems (CSC264)

Chapter-wise Study & Priority Guide

BSc. CSIT – 4th Semester – Tribhuvan University

Based on TU Exam Papers: 2076, 2078, 2079, 2080(New), 2081, 2082, Model (2077 & original 2080 papers not available at time of analysis)

■ ×1 Low – appeared once	■ ×2 Medium – appeared twice
■ ×3–4 High priority	■ ×5+ MUST PREPARE

★ = extremely high priority — near-guaranteed numerical or core-concept question, practice by hand.

Unit 1 – Operating System Overview (4 Hrs)

System Call & Handling System Calls ■ ×4 ★

Years: 2078, 2079, 2080(New), Model

Q: Define system call; describe how system calls are handled, including process state transitions triggered during the call.

OS Structures & Kernel Models (microkernel, exokernel, open-source OS, OS modes) ■ ×3

Years: 2076, 2081, 2082

Q: Explain microkernels and exokernels; discuss open-source operating systems and the two modes of OS operation.

Two Main Functions / Views of an Operating System ■ ×1

Years: 2082

Q: List and describe the two main functions of an operating system.

■ **What to read:** Short unit, definitional in nature — system call handling is the recurring anchor; know the flow (user mode → trap → kernel mode → return) cold. OS structure comparisons (micro/exo/monolithic) are quick 5-mark answers, don't over-invest time here.

Unit 2 – Process Management (10 Hrs)

CPU Scheduling Numerical (FCFS / SJF / SRTN / Round Robin / Priority — avg waiting & turnaround time) ■ ×6 ★

Years: 2076, 2078, 2079, 2080(New), 2081, 2082

Q: Compute average waiting time and turnaround time for a given set of processes using two or more scheduling algorithms (FCFS, SJF, SRTN, RR with given quantum, Priority).

Mutual Exclusion Techniques (busy waiting, lock variable, disabling interrupts, TSL, Peterson's solution) ■ ×4 ★

Years: 2078, 2079, 2080(New), 2082

Q: Explain how a given mutual exclusion technique (lock variable / busy wait / disabling interrupts) works and its associated problems.

Classical IPC Problems (Sleeping Barber, Producer-Consumer, Sleep & Wakeup) ■ ×4

Years: 2078, 2079, 2081, Model

Q: Describe a classical IPC problem (e.g. Sleeping Barber, Sleep and Wakeup) and illustrate how it is solved, often with a code snippet.

Race Condition & Critical Section ■ ×4

Years: 2079, 2081, 2082, Model

Q: Define race condition and critical section; justify how a race condition can lead to data loss or incorrect data.

Semaphores & Monitors ■ ×2

Years: 2076, 2081

Q: Explain how semaphores solve the critical section problem; discuss the concept of monitors and problems associated with semaphores.

Threads vs Processes ■ x2

Years: 2076, Model

Q: Differentiate threads from processes; explain when threads are preferable and describe user-level threads.

Scheduling Concepts (goals, time quantum effects, multiple queues, preemptive vs non-preemptive) ■ x3

Years: 2079, 2082, Model

Q: Discuss the goals of interactive system scheduling; explain why multiple queues are needed and how time quantum affects Round Robin performance.

■ **What to read:** This is the highest-weight unit (10 hrs) and it shows in the paper. CPU scheduling numericals are a guaranteed question every year — drill Gantt-chart-style calculations for all five algorithms until you're fast. Pair mutual exclusion + IPC problems + race condition together since they're conceptually linked.

Unit 3 – Process Deadlocks (6 Hrs)

Deadlock vs Starvation vs Livelock (conceptual differentiation) ■ ×5 ★

Years: 2076, 2079, 2080(New), 2082, Model

Q: Differentiate between deadlock, starvation and livelock, often alongside a scenario-based follow-up question.

Banker's Algorithm / Safe State Numerical ■ ×4 ★

Years: 2078, 2079, 2082, Model

Q: Given allocation, maximum need and available resource matrices, determine if the system is in a safe state, compute the Need matrix, or find the smallest resource value for a safe state.

Deadlock Avoidance & Safe/Unsafe State ■ ×2

Years: 2080(New), 2081

Q: Illustrate the terms safe and unsafe state in deadlock prevention/avoidance with a scenario; explain how deadlock can be avoided.

Resource Allocation Graph & Deadlock Detection ■ ×2

Years: 2076, 2078

Q: Explain the Resource Allocation Graph and the process of detecting deadlocks for single and multiple resource instances.

■ **What to read:** Banker's Algorithm is the numerical to master — practice computing Need, checking safe sequences, and evaluating whether a resource request can be granted immediately. The deadlock/starvation/livelock distinction is asked almost every year as a lead-in before the numerical, so have a crisp one-line definition ready.

Unit 4 – Memory Management (8 Hrs)

Page Replacement Algorithm Numerical (FIFO / LRU / Optimal / Second Chance / Clock) ■ ×6

★

Years: 2076, 2078, 2079, 2080(New), 2082, Model

Q: Given a page reference string and number of frames, compute the number of page faults using two or more page replacement algorithms (FIFO, LRU, Optimal, Second Chance, Clock).

Segmentation (concept, address translation, drawbacks) ■ ×3

Years: 2079, 2081, 2082

Q: Define segmentation, explain the translation of logical to physical address using a segment table with a diagram, and list its advantages/disadvantages.

Virtual Memory / Paging / Page Table / TLB ■ ×3

Years: 2079, 2080(New), 2081

Q: Explain why virtual memory is needed, describe the structure of a page table, and explain the working mechanism and advantages of the TLB.

Memory Address Translation Numerical (segment table lookups, virtual address → page number/offset) ■ ×3

Years: 2081, 2082, Model

Q: Given a segment table or page size, compute the physical address for given logical addresses, or the page number and offset for given virtual addresses.

Memory Partitioning Numerical (First Fit / Next Fit / Best Fit / Worst Fit) ■ ×2

Years: 2078, 2081

Q: Given a list of memory hole sizes and incoming process requests, determine which partition is allocated under First Fit, Next Fit, Best Fit and Worst Fit.

Belady's Anomaly ■ ×2

Years: 2076, 2082

Q: Explain Belady's Anomaly with a suitable example, typically alongside a page replacement comparison.

Fragmentation (internal vs external) ■ ×2

Years: 2078, Model

Q: Differentiate internal fragmentation from external fragmentation.

Working Set & Locality of Reference ■ x2

Years: 2079, 2080(New)

Q: Define working set and explain why locality of reference matters; describe how the Clock replacement algorithm works.

■ **What to read:** Second heaviest numerical unit after scheduling. Page replacement questions (FIFO/LRU/Optimal) appear in 6 of 7 papers — practice the frame-table trace method until you can do it without mistakes under time pressure. Memory partitioning and address translation numericals are shorter but still frequent — don't skip them.

Unit 5 – File Management (6 Hrs)

File Allocation Methods (contiguous, linked list, i-nodes, FAT) ■ x5 ★

Years: 2078, 2079, 2080(New), 2082, Model

Q: Explain how files are implemented using a given allocation method (linked list, contiguous, i-nodes); compute disk space usage or number of disk operations for a given scenario.

Directory Systems (single/two-level/hierarchical, implementation techniques) ■ x4

Years: 2078, 2079, 2080(New), Model

Q: Discuss single-level, two-level and hierarchical directory systems; explain a directory implementation technique in detail.

Free Space Management ■ x1

Years: 2078

Q: Explain approaches for managing free disk space, particularly the linked list approach, with an example.

■ **What to read:** I-nodes and linked-list file allocation are the recurring numerical/conceptual anchor — make sure you can compute disk operations or space overhead, not just describe the method. Directory systems are usually a clean definitional question — low effort, consistent marks.

Unit 6 – Device Management (6 Hrs)

Disk Scheduling Numerical (FCFS / SSTF / SCAN / C-SCAN / LOOK / C-LOOK) ■ x5 ★

Years: 2076, 2079, 2080(New), 2081, 2082

Q: Given a disk request queue, initial head position and cylinder range, compute total seek time / head movement using two or more disk scheduling algorithms.

DMA, Interrupts & I/O Handling ■ x3

Years: 2076, 2079, 2080(New)

Q: Describe how DMA operation is performed; explain programmed I/O vs interrupt-driven I/O vs DMA-based I/O.

RAID / Disk Formatting / Interleaving ■ x3

Years: 2078, 2080(New), 2082

Q: Explain the concept of disk interleaving and cylinder skew; write short notes on RAID.

Memory-Mapped I/O ■ x1

Years: 2081

Q: Explain memory-mapped I/O.

■ **What to read:** Disk scheduling numericals are close to guaranteed — practice SCAN/C-SCAN/LOOK/C-LOOK carefully since they're easy to mix up under pressure (know exactly when the head reverses direction for each). DMA and RAID are safer definitional/short-note territory.

Unit 7 – Linux Case Study (5 Hrs)

IPC in Linux / Linux Process Management / Signals ■ x3

Years: 2078, 2081, 2082

Q: Explain Inter-Process Communication mechanisms in Linux; discuss why Linux restricts signals to processes within the same process group.

Linux Scheduling & Linux File System ■ x2

Years: 2076, Model

Q: Write a short note on Linux scheduling or the Linux file system.

■ **What to read:** Smallest unit — usually shows up as a short note (Q12) rather than a full question. Know Linux IPC mechanisms (pipes, signals, message queues) at a definitional level; don't over-study this relative to Units 2 and 4.

Top Priority Summary (Sorted by Frequency)

Topic	Unit	Freq	Years
CPU Scheduling Numerical (FCFS/SJF/SRTN/RR/Priority)	Process Management	×6	2076–82
Page Replacement Algorithm Numerical (FIFO/LRU/OPR)	Memory Management	×6	2076–82,Model
Deadlock vs Starvation vs Livelock	Process Deadlocks	×5	2076,79,80(N),82,Model
File Allocation Methods (linked list/i-node/FAT)	File Management	×5	2078,79,80(N),82,Model
Disk Scheduling Numerical (SCAN/SSTF/LOOK etc.)	Device Management	×5	2076,79,80(N),81,82
Mutual Exclusion Techniques	Process Management	×4	2078,79,80(N),82
Classical IPC Problems	Process Management	×4	2078,79,81,Model
Race Condition & Critical Section	Process Management	×4	2079,81,82,Model
Banker's Algorithm / Safe State Numerical	Process Deadlocks	×4	2078,79,82,Model
Directory Systems (levels & implementation)	File Management	×4	2078,79,80(N),Model
System Call & Handling	OS Overview	×4	2078,79,80(N),Model
OS Structures & Kernel Models	OS Overview	×3	2076,81,82
Scheduling Concepts (goals, quantum, queues)	Process Management	×3	2079,82,Model
Segmentation	Memory Management	×3	2079,81,82
Virtual Memory / Paging / TLB	Memory Management	×3	2079,80(N),81
Memory Address Translation Numerical	Memory Management	×3	2081,82,Model
DMA, Interrupts & I/O Handling	Device Management	×3	2076,79,80(N)
RAID / Disk Formatting / Interleaving	Device Management	×3	2078,80(N),82
IPC in Linux / Process Mgmt	Linux Case Study	×3	2078,81,82

Final Exam Very High Probability Topics (Never Miss These)

MUST PREPARE (×5+, near-certain): CPU scheduling numerical (all algorithms), Page replacement numerical (FIFO/LRU/OPR), Deadlock/starvation/livelock differentiation, File allocation methods, Disk scheduling numerical.

High priority (×3–4): Mutual exclusion techniques, Classical IPC problems, Race condition, Banker's Algorithm, Directory systems, System calls, Segmentation, Virtual memory/TLB, Memory address translation.

Numerical-heavy (practice by hand every time): Scheduling avg-waiting/turnaround tables, page-fault frame traces, Banker's safe-state check, disk-seek-time calculations, memory partition allocation.

Due for a repeat (asked before, quiet recently — good bet): Memory-mapped I/O, Working set/locality of reference, Belady's anomaly, Threads vs processes, Fragmentation, RAID.